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Probing Feature Universality in Medical Vision Models

by

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Preface

The research performed in this thesis was done in collaboration with Nikolay L. Tsonev; Section 3.2, Section 3.3 and Section 3.5 have been worked on together.

Abstract

Medical AI systems that match specialist performance on diagnostic benchmarks are increasingly deployed in clinical workflows, yet the internal reasoning behind their decisions remains largely opaque. A model that cannot account for the patterns driving its predictions cannot be reliably trusted when patient populations, imaging hardware, or disease prevalence shift from those seen during training. Interpreting what these models have actually learned is therefore a prerequisite for assessing their robustness.

This thesis asks whether a multi-task medical foundation model, UMedPT, encodes stable, clinically meaningful internal features that recur consistently across different patient populations and disease contexts. To answer this, a technique from the emerging field of mechanistic interpretability is applied: a sparse autoencoder is trained to decompose the model's internal activations into a dictionary of individual features, each associated with a specific and interpretable concept.

The investigation extracts activations from the model's final pre-pooling representation stage across four chest X-ray datasets spanning distinct patient populations, institutions, and disease categories. A systematic evaluation pipeline then tests each discovered feature for cross-dataset consistency, acquisition independence, spatial anatomical alignment, and semantic coherence.

Of 4,096 candidate features, 65 are confirmed as universal: they fire consistently across two of the largest datasets, align in the model's representational space, and a subset localise to anatomically meaningful image regions confirmed by both radiologist bounding-box annotations (where available) and manual inspection of activation maps across all four datasets. These results indicate that mechanistic interpretability methods developed for language models extend to medical vision models, and that a sparse linear decomposition of UMedPT's activations recovers cross-dataset clinical structure beyond what either linear dimensionality reduction or random projection accounts for.

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1 Introduction

A medical vision model that classifies a chest radiograph as showing consolidation provides only a label, and not a verifiable account of which patterns drove the decision or if those patterns correspond to clinically meaningful concepts [33]. This is the opacity that limits the robustness of medical AI systems: a model may exploit spurious correlations that hold within a training distribution but break when patient populations, acquisition hardware, or disease prevalence shift. Interpretability methods that reveal *what* a model has actually learned therefore have direct value for assessing and improving the reliability of medical imaging models [46].

Mechanistic interpretability (MI) addresses this by treating a trained neural network, with the goal of decomposing its computations into human-understandable components [5, 27]. Applied to large language models (LLMs), MI has produced detailed accounts of circuits for specific reasoning tasks [11], features encoding factual knowledge [45], and a systematic understanding of why individual neurons encode multiple unrelated concepts simultaneously – a phenomenon termed *polysemanticity* arising from *superposition* [12]. Sparse autoencoders (SAEs) have emerged as the primary tool for resolving superposition: by enforcing sparsity over an overcomplete dictionary, they decompose a model's activations into *monosemantic* features, each encoding a single interpretable concept [7, 13]. This line of work has concentrated primarily on language models, with limited application to medical vision models [33].

This thesis applies SAE-based mechanistic interpretability to a medical foundation model. The model studied is UMedPT [38], a Swin Transformer architecture [24] pre-trained jointly on a diverse corpus of medical imaging tasks. In this thesis, the model is evaluated exclusively on chest radiography; its multi-task pre-training is exploited as a regulariser promoting general visual representations. This training regime makes UMedPT a strong candidate for having developed *universal* internal representations – shared feature encodings that reflect candidate clinical concepts rather than dataset-specific acquisition artefacts. The central question here is:

> *Are there pathological and anatomical concepts encoded in the internal representations of UMedPT that are consistent across patients, datasets, and disease contexts within the chest radiography domain?*

This question operationalises the *universality hypothesis* from the circuits literature [30], which proposes that analogous features form in neural networks trained on similar domains. The original hypothesis was formulated for vision CNNs and tested across model architectures. This thesis redefines and applies it within a single foundation model, testing cross-dataset consistency. Within-modality universality, asking whether a single model's features generalise across patient populations, institutions, and disease contexts while remaining fixed in architecture is a natural and necessary first step before any cross-model or cross-modality claim. The scope is limited accordingly: universality is operationalised as cross-dataset agreement on activation frequency and representational direction, evaluated primarily on the two largest datasets (CheXpert and RSNA Pneumonia); the smaller SIIM-COVID19 and TB Shenzhen corpora serve as auxiliary evidence, as their token counts fall below the frequency threshold for all confirmed latents.

The approach is to train a BatchTopK SAE [8] on activations extracted from Medical Vision Model, UMedPT, across four chest X-ray datasets spanning distinct patient populations, institutions, and disease contexts: CheXpert [16], RSNA Pneumonia [41], SIIM-

COVID19 [43], and the TB Shenzhen corpus [17]. The SAE's dictionary atoms are then evaluated through a hierarchical universality pipeline, yielding a graded characterisation of which latents encode stable cross-dataset clinical concepts.

This thesis makes the following contributions:

1. **A hierarchical operationalisation of feature universality for SAE latents.** A latent is counted as universal if it passes the two core gates: (i) activation frequency above 1% on at least two of four datasets, and (ii) representational direction alignment (cosine similarity ≥ 0.70 between per-dataset activation centroids in d_{model} space). Two further criteria are reported as supplementary characterisation rather than as required gates: (iii) spatial grounding against radiologist bounding boxes (Bonferroni-corrected Pearson r pooled across images), which separates the spatially grounded subset from the broader confirmed set; and (iv) monosemanticity, assessed by Jaccard co-activation split detection.
2. **A scanner-latent audit pipeline.** Before universality analysis, latents encoding acquisition differences rather than pathology are identified via multinomial logistic regression predicting dataset identity from per-latent activation magnitudes (macro AUROC $> 0.80 \Rightarrow$ excluded), guarding against confounding scanner signatures with clinical concepts.
3. **An illness-stratified clinical characterisation of confirmed universal latents.** Universal features are classified along a gradient from geometric framing detectors to pathology-presence detectors using a pathology-preference ratio ρ_k and cross-validated discriminability probes (macro OvR AUROC, 5-fold stratified CV).
4. **Null baseline validation.** The full universality pipeline is repeated on a randomly initialised SAE of the same architecture and on a PCA baseline, establishing that confirmed universal latents in the trained SAE exceed what is expected by chance or by linear dimensionality reduction.

Section 2 reviews the literature on mechanistic interpretability, sparse autoencoders, and medical vision explainability. Section 3 describes the investigation: dataset selection, the activation harvesting pipeline, the BatchTopK SAE architecture and training procedure, and the full universality evaluation pipeline. Section 4 presents and evaluates the results, including a quantitative analysis of confirmed universal latents, comparison against null baselines, and illness-stratified characterisation. Section 5 concludes with a direct answer to the research question and directions for future work.

2 Statement and Motivation of Research

2.1 Motivation

The research question stated in Section 1 is motivated by two premises: (1) Whether a medical vision model encodes clinical concepts or exploits training-specific regularities cannot be determined by measuring performance alone. Interpretability methods that characterise internal representations provide a deeper account of what the model has learned [46]. (2) Multi-task foundation models trained on heterogeneous data are expected, under the universality hypothesis [30], to develop shared internal representations for concepts that appear consistently across their training distribution. If such a model has learned *universal* representations, this would suggest that the model's pre-training captured features stable to acquisition and cohort variation, rather than dataset-specific artefacts.

2.2 Background

2.2.1 Mechanistic Interpretability

Modern neural networks are typically described by their architecture and training objective, but not by the internal computations they implement [30]. Mechanistic interpretability (MI) aims to recover a human-understandable description of a trained network's computations from its weights and activations [5, 27]. The goal is not explanation-after-the-fact but *reverse engineering*: to identify the components of a network – features, circuits, attention patterns – and to characterise their functional roles in terms that correspond to meaningful concepts.

A central challenge for MI is *polysemanticity*: individual neurons in trained networks encode multiple unrelated concepts simultaneously [7]. A single neuron in a vision model may respond to unrelated visual patterns across images; a neuron in a language model may encode semantically unrelated words. Polysemanticity makes direct neuron-level inspection unreliable as an interpretability strategy.

Elhage et al. [12] provide a formal explanation for polysemanticity in terms of *superposition*. A network layer with d_{model} neurons can, under sparsity conditions, represent more than d_{model} features simultaneously by distributing each feature across multiple neurons and encoding features in directions that are nearly orthogonal in the high-dimensional activation space. Superposition is optimal when features are sparse – present only occasionally – which is true of most structured concepts (a particular pathology, a grammatical construction) in natural data. The consequence is that the natural basis of a network's representations is not aligned with the neuron basis: features are encoded as directions in activation space, not as individual neurons.

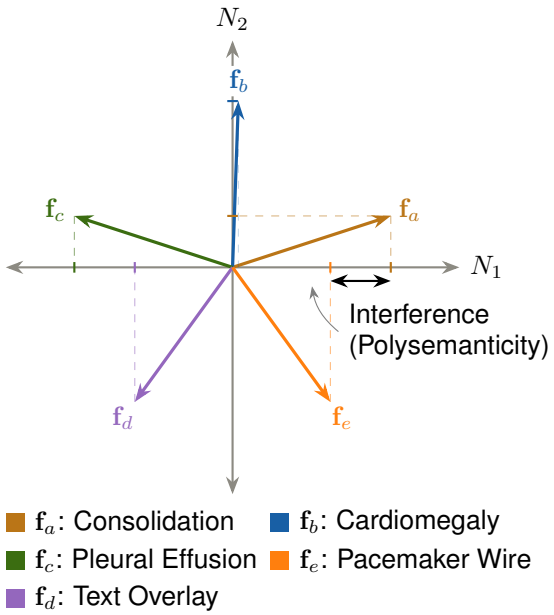


Figure 1: **Geometric representation of superposition [12].**: Features $\mathbf{f}_a$–$\mathbf{f}_e$ are encoded as directions in a two-neuron space (N_1, N_2).

MI has produced detailed accounts of specific computations in language models. Elhage et al. [11] introduced the transformer circuits framework, decomposing attention patterns into interpretable operations on residual stream vectors. Nanda et al. [26] used MI to track the internal mechanisms by which transformers undergo grokking; a delayed generalisation phenomenon. Templeton et al. [45] extracted thousands of interpretable features from Claude 3 Sonnet using sparse autoencoders. A comprehensive review of the field's scope and open problems is given in [5, 40].

2.2.2 Sparse Autoencoders

Sparse autoencoders (SAEs) have become the primary tool for resolving superposition and recovering monosemantic features from polysemantic neuron activations. An SAE is trained to reconstruct a model's activation vector $\mathbf{x} \in \mathbb{R}^{d_{\text{model}}}$ via a sparse intermediate code $\mathbf{z} \in \mathbb{R}^{d_{\text{SAE}}}$ with $d_{\text{SAE}} \gg d_{\text{model}}$:

$$\mathbf{z} = \text{TopK}(\text{ReLU}((\mathbf{x} - \mathbf{b}_{\text{dec}})\,\mathbf{W}_{\text{enc}} + \mathbf{b}_{\text{enc}}))\,, \tag{1}$$

$$\hat{\mathbf{x}} = \mathbf{z}\,\mathbf{W}_{\text{dec}} + \mathbf{b}_{\text{dec}} \tag{2}$$

The BatchTopK variant adopted in this thesis replaces the per-sample TopK with a batch-level top-$k \cdot B$ selection; the resulting formulation is given in Section 3.4.

The overcomplete dictionary $\mathbf{W}_{\text{dec}} \in \mathbb{R}^{d_{\text{SAE}} \times d_{\text{model}}}$ contains d_{SAE} learned feature directions (*dictionary atoms*), each constrained to unit norm. Sparsity forces the SAE to allocate separate dictionary atoms to separate concepts, resolving the superposition that distributes each concept across multiple neurons.

Early work by Bricken et al. [7] demonstrated that SAE features trained on one-layer MLP activations in language models are monosemantic and interpretable at scale. Their training

used an L1 penalty to enforce sparsity, leading to *shrinkage*: learned activations are systematically underestimated because L1 penalises magnitude uniformly. Rajamanoharan et al. [36] addressed shrinkage through Gated SAEs, which separate the gate (feature presence) and magnitude (feature value) estimation paths. JumpReLU SAEs [37] replace the L1 penalty with a learnable threshold, directly targeting the L_0 sparsity count via straight-through gradient estimation, achieving state-of-the-art reconstruction on Gemma 2 9B.

Gao et al. [13] introduced TopK SAEs, which replace the L1 penalty with a hard top-k constraint: exactly k features are active per token. This eliminates shrinkage by construction, yields clean scaling laws for reconstruction quality as a function of d_{SAE} and k, and simplifies evaluation because the sparsity level is a fixed parameter rather than an emergent property of training. Gemma Scope [23] applied this approach to train over 400 JumpReLU SAEs across all layers of Gemma 2, releasing open SAE weights to support community MI research. A comprehensive survey of SAE architectures, training strategies, and evaluations is given in [42].

Bussmann et al. [8] introduced BatchTopK SAEs, which relax the per-sample top-k constraint to a batch-level constraint: exactly $k \times B$ features are active across a batch of B tokens, rather than exactly k per token. This allows the allocation of active features to adapt to sample complexity – tokens corresponding to pathology-rich images receive more active features than anatomically simple images – while maintaining the same average sparsity as TopK over the training distribution. BatchTopK matches or exceeds JumpReLU performance on reconstruction benchmarks while preserving the training stability of hard-constraint approaches.

End-to-end SAEs [6] extend this framework by training with a KL divergence objective against the model's output distribution rather than a reconstruction loss, recovering features that are functionally important to model behaviour rather than those that maximally explain activation variance. This approach is not used in this thesis; the standard reconstruction-based objective is sufficient for the universality analysis, which operates on activation structure rather than model outputs.

2.2.3 Circuits and the Universality Hypothesis

Olah et al. [29] introduced the circuits framework for interpretable analysis of vision models, demonstrating that CNNs can be partially reverse-engineered by identifying interpretable features (neurons or linear combinations thereof) and the computational circuits that connect them. Feature visualisation [28] showed that early convolutional layers learn Gabor-like filters and colour detectors; intermediate layers assemble curve detectors and texture responses; late layers encode object parts and scene compositions.

In the circuits thread, Olah et al. [30] stated three claims that underpin the universality hypothesis: (1) features are the natural unit of neural network representations; (2) features connect into circuits implementing meaningful computations; and (3) features and circuits are *universal* – analogous features form independently across models and architectures trained on similar domains. Supporting evidence included Gabor filters and curve detectors found in independently trained CNNs with different architectures. Cammarata et al. [9] provided a detailed case study of curve detector circuits in InceptionV1.

Cross-architecture representational similarity has been studied quantitatively using metrics such as Centred Kernel Alignment (CKA) [22], which measures the alignment of

representational geometry between layers of independently trained networks. High CKA between corresponding layers of different architectures supports the claim that similar representational structure emerges from similar training objectives. The cross-architecture universality question – whether distinct models converge to similar internal solutions – is a distinct and prior question from the within-model cross-dataset question addressed in this thesis. This thesis does not test cross-architecture universality.

2.2.4 Vision SAEs and the Medical Interpretability Gap

MI and SAE analysis have been applied predominantly to language models, with extension to vision transformers a recent development. Initial work demonstrated that monosemantic features emerge in fine-tuned ViTs [3] and that SAE decomposition of CLIP's vision encoder isolates semantic rather than purely visual concepts [20]. Subsequent studies established that these features are not merely correlational: patch-level intervention on a frozen ViT alters classification and segmentation outputs in predictable ways [44], and ablation-based analysis recovers functional roles that activation maps alone misrepresent [14]. Comparable monosemanticity has been observed in vision-language models [31], indicating that the SAE approach transfers across both unimodal and multimodal vision architectures.

In medical imaging, by contrast, interpretability has remained at the level of post-hoc attribution. Gradient-based saliency [39], attention rollout, and perturbation methods produce pixel-level importance maps for individual predictions but do not characterise the internal representations that generate them; the dominant survey in this domain [33] and ViT-specific benchmarks [4] both confirm that attribution remains the prevailing paradigm. Saliency methods have moreover been shown to be insensitive to model and data randomisation, undermining their reliability as explanations [1]. Patrício et al. [33] accordingly identify mechanistic interpretability as an open direction in medical imaging. The natural target is a medical vision foundation model: UMedPT [38], a Swin Transformer [24] pretrained jointly on classification, segmentation, and detection across multiple modalities, provides a frozen encoder applicable to any medical image without task-specific fine-tuning. Unlike the flat Vision Transformer, which retains a single token grid throughout [10], the Swin Transformer is *hierarchical*: it processes the image in four stages, halving the spatial resolution at each PatchMerging boundary while doubling the channel dimension, so that successively deeper stages encode progressively more abstract features. Self-attention is computed within shifted local windows rather than globally, which keeps the cost linear in the number of tokens and allows the architecture to scale to the high-resolution inputs typical of medical imaging. The property relevant to this thesis is that each stage exposes a clean activation tensor of known shape, for UMedPT, $7 \times 7 \times 1024$ at the output of Stage 4, providing a discrete extraction point at which an SAE can be applied without the architectural ambiguity of choosing a layer index inside a flat ViT. To our knowledge, no prior work applies SAE-based mechanistic interpretability to such a model, nor tests the universality hypothesis within medical imaging. Existing cross-architecture similarity metrics such as CKA [22] operate on representational geometry; this thesis instead asks whether individual dictionary atoms – features with specific learned directions – fire consistently across datasets, a finer-grained and more interpretable test.

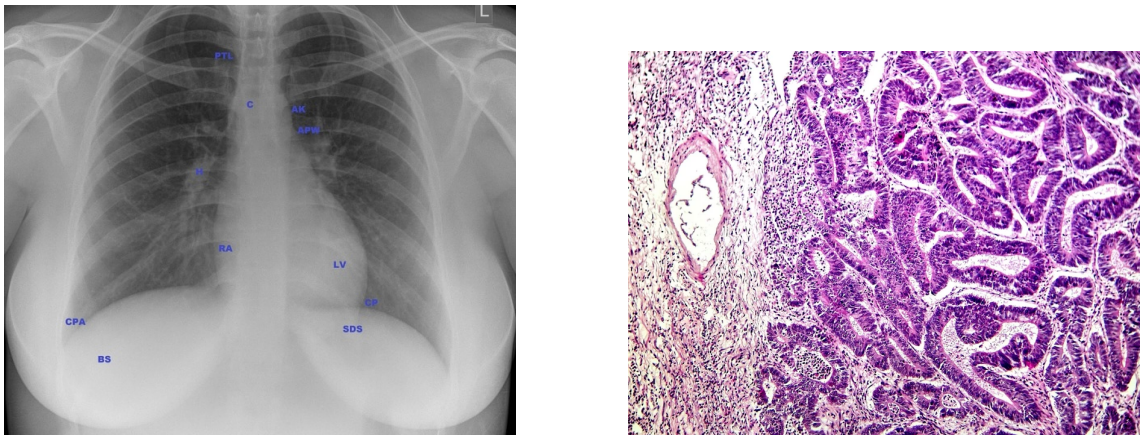
3 Description of the Investigation

3.1 Overview

The investigation proceeds in three phases. Phase 1 extracts activations from a frozen UMedPT encoder applied to four chest X-ray datasets, producing a pooled activation corpus. Phase 2 trains a BatchTopK SAE on this corpus, learning a sparse overcomplete dictionary of features from the encoder's representations. Phase 3 evaluates the trained SAE through a hierarchical universality pipeline, characterising which latents encode stable cross-dataset clinical concepts, which encode scanner or acquisition signatures, and which are pathology-discriminative or spatially localised.

3.2 Modality and Datasets

Chest radiography was chosen as the evaluation modality on interpretability grounds. Qualitative adjudication (Section 3.6, Step 7) requires that a latent's top-activating patches be visually recognisable as anatomical or pathological structure. Chest X-rays satisfy this at the resolution and projection geometry a non-specialist can read; histopathology slides of colorectal or breast tissue (Figure 2) carry their diagnostic signal at cellular scale and demand pathology training to interpret.



(a) Chest radiograph

(b) Histopathology slide

Figure 2: **Modality comparison.**

Within chest radiography, four datasets were selected to vary the axes along which universality is non-trivial – institution, geography, label scheme, and dominant disease – so that any feature confirmed across them has resisted every shift the construct is meant to capture. Table 1 summarises their composition as used in this thesis.

Dataset	Images	Labels	BBoxes	View
CheXpert [16]	10,000	14 binary $(-1 \to 0)$	No	PA
RSNA Pneumonia [41]	10,000	Binary pneumonia	Yes	PA
SIIM-COVID19 [43]	6,334	4-class study-level	Yes	PA
TB Shenzhen [17]	662	Binary TB/normal	No	Frontal

Table 1: Chest X-ray datasets. "Images" is the sampled count, not the full dataset size. BBoxes = radiologist bounding boxes available.

CheXpert is a large-scale chest radiograph dataset from Stanford Medical Center with 14 pathology labels annotated by an automated labeller from radiology reports, including explicit uncertainty annotations. **RSNA Pneumonia** (Kaggle 2018 challenge) provides binary pneumonia labels and radiologist bounding boxes for positive cases, drawn from a general US hospital population. **SIIM-COVID19** (Kaggle 2021 challenge) provides four-class opacity labels (negative, atypical, indeterminate, typical) with bounding boxes, covering an international COVID-19 patient cohort. **TB Shenzhen** provides binary tuberculosis labels from a Chinese hospital cohort, representing a substantially different demographic and disease distribution from the Western datasets. Because TB Shenzhen contains only 662 images, each image is presented approximately 14 times per epoch under stratified sampling. To reduce the risk of the SAE memorising specific image instances under this oversampling, random augmentation was applied at harvest time: horizontal flip ($p = 0.5$), brightness jitter ($\pm 10\%$), and random crop (scale 0.85–1.00) with resize back to 224×224. No augmentation was applied to the three larger datasets.

Where dataset metadata permitted, AP/PA view filtering was applied during dataset loading to reduce the Anteroposterior (AP) / Posteroanterior (PA) geometric confound: portable AP examinations systematically magnify cardiac silhouette size relative to PA standing examinations [25], producing acquisition-dependent features irrelevant to pathology. CheXpert was filtered strictly to PA views by requiring both `Frontal/Lateral` = `Frontal` and `AP/PA` = `PA` in the training CSV; lateral and AP views were excluded. SIIM-COVID19 was filtered where view metadata was present in the image-level CSV; rows without a `view` field were retained, as the challenge dataset is predominantly PA by design. RSNA Pneumonia and TB Shenzhen were not filtered by view metadata: the RSNA challenge training set is drawn from PA examinations by dataset convention, and the TB Shenzhen images are all frontal chest radiographs.

Pathology labels across the four datasets use incompatible label schemes. Before any label-conditioned analysis, raw labels are mapped to 7 canonical groups following the Fleischner Society Glossary of Terms for Thoracic Imaging [15]: `airspace_opacity`, `pleural_abnormality`, `cardiac_mediastinal`, `atelectasis`, `vascular_edema`, `no_finding`, and `other`. This taxonomy is also the labelling convention used by CheXpert. Inter-rater agreement for raw chest X-ray labels has been reported at $\kappa = 0.53$–0.68 [18], making direct raw-label matching across datasets too strict; the Fleischner grouping provides a clinically grounded common vocabulary.

3.3 Activation Harvesting

3.3.1 Encoder

UMedPT was selected for the breadth of its pretraining along the two axes relevant to a universality study: tasks (classification, segmentation, detection) and modalities (radiography, histopathology, ophthalmology, dermatology, ultrasound). Features that survive this range of training objectives are unlikely to be idiosyncratic to any one of them, raising the prior that surviving cross-dataset features reflect genuine clinical structure rather than a single task signal. Its Swin Transformer backbone additionally exposes discrete stage boundaries for activation extraction.

The UMedPT encoder (release version 1.0.4) was used as released, with all weights frozen throughout. No fine-tuning on the chest X-ray datasets was performed. Activations were extracted in a single forward pass per image with the encoder in evaluation mode,

with gradients disabled. The frozen encoder was applied to all images in the four datasets in a single harvesting pass, storing activations, dataset identity labels, and image-stem metadata in a single merged `.safetensors` file.

Using the pre-trained (not task-fine-tuned) encoder is a deliberate design choice. UMedPT provides separate task-specific fine-tuned checkpoints for individual downstream tasks. A fine-tuned checkpoint would introduce task-specific adaptation shifts into the activation distributions, making cross-dataset comparisons confounded by the fine-tuning objective rather than the pre-training representations. This frozen choice ensures that any universality observed in the SAE latents is attributable to the foundation model's pre-training rather than task-specific adaptation.

3.3.2 Extraction Stage

Activations were extracted at Stage 4 of the UMedPT Swin Transformer, capturing the output of Stage 4's transformer blocks: the final representations produced by the backbone before global average pooling and the classification head. At this stage the spatial resolution is $7 \times 7 = 49$ tokens per image and the feature dimension is $d_{\text{model}} = 1024$. These two extraction parameters are identical to the Stage 3→4 boundary, which captures the PatchMerging output between Swin Stages 3 and 4 – representations after spatial downsampling to 7×7 but before Stage 4's transformer blocks integrate global context at this resolution.

Because both stages share identical dimensionality and spatial resolution, the difference between them isolates the contribution of Stage 4's self-attention integration. Stage 4 was selected: applying the universality pipeline under identical protocol confirmed 65 universal latents at Stage 4 versus 45 at Stage 3→4 – a 44% increase with the same SAE dictionary size. This is consistent with evidence that final-stage transformer representations encode more semantically integrated features than transition-point representations [34, 32].

3.3.3 Normalisation Pipeline

Three normalisation steps are applied in sequence to remove confounds at different levels of the activation hierarchy.

RMS normalisation. Following Gao et al. [13], each token vector $\mathbf{x} \in \mathbb{R}^{d_{\text{model}}}$ is normalised to unit RMS and rescaled to have RMS $= \sqrt{d_{\text{model}}}$:

$$\tilde{\mathbf{x}} = \frac{\mathbf{x}}{\sqrt{\langle \mathbf{x}^2 \rangle + \varepsilon}} \cdot \sqrt{d_{\text{model}}} \tag{3}$$

This removes per-token magnitude differences caused by X-ray exposure settings (kVp, mAs) and patient body habitus, which produce activation vectors of systematically different scales across images from different hospitals. The rescaling factor $\sqrt{d_{\text{model}}}$ aligns the expected activation norm with the SAE weight initialisation, following the Anthropic convention [45].

Per-dataset mean centering. After all tokens for each dataset are harvested and RMS-normalised, the per-dataset mean activation vector is subtracted:

$$\bar{\mathbf{x}}_d = \mathbf{x}_d - \frac{1}{N_d} \sum_{n=1}^{N_d} \mathbf{x}_{d,n} \tag{4}$$

Different hospitals and scanners produce activation distributions centred at different points in $\mathbb{R}^{d_{\text{model}}}$, reflecting systematic directional biases from preprocessing pipelines, DICOM windowing, and scanner characteristics. Without centering, the SAE decoder bias $\mathbf{b}_{\text{dec}}$ drifts toward the dataset with the largest mean norm, causing systematic reconstruction error on all other datasets. After centering, all four datasets are centred near the origin and $\mathbf{b}_{\text{dec}}$ is initialised to zero.

Decoder unit-norm constraint. At each training step, the rows of the decoder weight matrix $\mathbf{W}_{\text{dec}}$ are constrained to unit norm by projecting the gradient to be tangent to the unit sphere and renormalising the weights (Equation 7):

$$\hat{\mathbf{w}} = \mathbf{W}_{\text{dec}} / \|\mathbf{W}_{\text{dec}}\|_2 \tag{5}$$

$$\mathbf{g}_{\perp} = \mathbf{g} - (\mathbf{g} \cdot \hat{\mathbf{w}})\, \hat{\mathbf{w}} \tag{6}$$

$$\mathbf{W}_{\text{dec}} \leftarrow \frac{\mathbf{W}_{\text{dec}} - \eta\, \mathbf{g}_{\perp}}{\|\mathbf{W}_{\text{dec}} - \eta\, \mathbf{g}_{\perp}\|_2} \tag{7}$$

This eliminates scale ambiguity between encoder activations and decoder directions, ensuring that feature activation magnitudes are directly comparable across latents. The constraint is applied after the backward pass and before the optimiser step, projecting decoder gradients tangent to the unit sphere before renormalising the weights.

3.4 SAE Architecture

The motivation for an SAE follows directly from the superposition hypothesis [12]: The encoder learns a wider basis $d_{\text{SAE}} \gg d_{\text{model}}$ and a sparsity constraint forces each input to be reconstructed as a sum of a small number of dictionary atoms, recovering the latent features the encoder had compressed into superposition. The decoder is deliberately linear: the linear representation hypothesis [12] states that features are encoded as directions, so a linear decoder is the correct inverse.

The SAE follows the BatchTopK formulation of Bussmann et al. [8]. The unit of processing is a single Swin Stage 4 patch token $\mathbf{x} \in \mathbb{R}^{d_{\text{model}}}$, so one image contributes 49 tokens; in practice the SAE always operates on a tensor of B tokens $\mathbf{X} \in \mathbb{R}^{B \times d_{\text{model}}}$. Tokens are processed independently through $\mathbf{W}_{\text{enc}}$ and $\mathbf{W}_{\text{dec}}$; the only inter-token interaction is BatchTopK, which shares a sparsity budget of $k \cdot B$ across the batch. The forward pass is:

$$\mathbf{A} = \text{ReLU}((\mathbf{X} - \mathbf{b}_{\text{dec}})\, \mathbf{W}_{\text{enc}} + \mathbf{b}_{\text{enc}}) \tag{8}$$

$$\mathbf{Z} = \text{BatchTopK}(\mathbf{A}, k \cdot B) \tag{9}$$

$$\hat{\mathbf{X}} = \mathbf{Z}\, \mathbf{W}_{\text{dec}} + \mathbf{b}_{\text{dec}} \tag{10}$$

where $\mathbf{W}_{\text{enc}} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{SAE}}}$ and $\mathbf{W}_{\text{dec}} \in \mathbb{R}^{d_{\text{SAE}} \times d_{\text{model}}}$ are encoder and decoder weight matrices, $\mathbf{b}_{\text{enc}} \in \mathbb{R}^{d_{\text{SAE}}}$ and $\mathbf{b}_{\text{dec}} \in \mathbb{R}^{d_{\text{model}}}$ are bias vectors, and $\text{BatchTopK}(\mathbf{A}, k \cdot B)$ zeros all but the $k \cdot B$ largest elements of the flattened batch activation matrix.

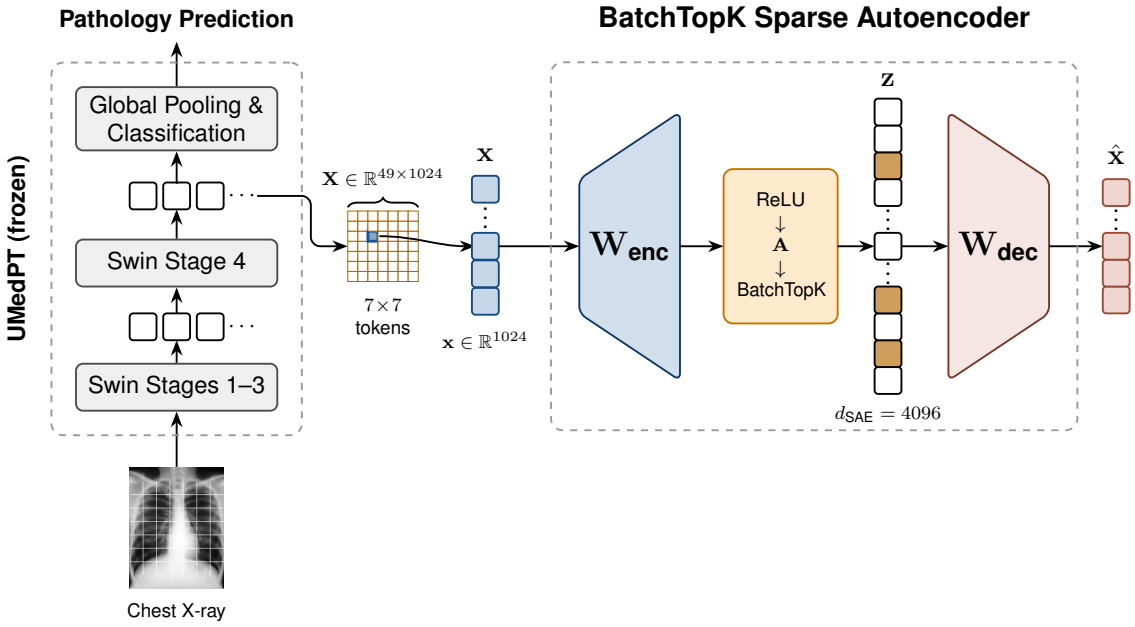


Figure 3: **BatchTopK Sparse Autoencoder architecture [8].** A chest X-ray is tokenised by the frozen UMedPT Swin backbone into $\mathbf{X} \in \mathbb{R}^{49\times1024}$ Stage 4 patch tokens. Each token $\mathbf{x} \in \mathbb{R}^{1024}$ (highlighted) is encoded independently via $\mathbf{W}_{\text{enc}}$, sparsified to a code $\mathbf{z} \in \mathbb{R}^{4096}$, and reconstructed as $\hat{\mathbf{x}}$ via $\mathbf{W}_{\text{dec}}$. Sparsity is enforced by BatchTopK during training (global top-$k\cdot B$ across the batch) and per-sample top-k at inference.

The BatchTopK constraint differs from per-sample TopK [13] in that it distributes the $k \cdot B$ active feature slots across the full batch rather than fixing exactly k per token. Tokens corresponding to pathology-rich or complex images receive more active features; tokens from simpler images receive fewer, with the same expected allocation of k per token across the batch. This relaxation is particularly appropriate for heterogeneous training data where image complexity varies systematically across datasets.

The training loss combines a reconstruction term with a dead-feature revival term; sparsity is enforced structurally by BatchTopK rather than through any explicit penalty:

$$\mathcal{L}_{\text{recon}} = \frac{1}{B\, d_{\text{model}}} \|\hat{\mathbf{X}} - \mathbf{X}\|_F^2, \tag{11}$$

$$\mathcal{L}_{\text{aux}} = \frac{1}{B\, d_{\text{model}}} \|\hat{\mathbf{X}}_{\text{aux}} - (\mathbf{X} - \hat{\mathbf{X}})\|_F^2, \tag{12}$$

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \alpha\, \lambda_{\text{aux}}\, \mathcal{L}_{\text{aux}}. \tag{13}$$

A feature is considered dead when it has been inactive for $\geq n_{\text{dead}}$ consecutive batches; let $\mathcal{D}$ denote the set of dead features at the current step. The auxiliary reconstruction $\hat{\mathbf{X}}_{\text{aux}} = \mathbf{Z}_{\text{aux}}\, \mathbf{W}_{\text{dec}}^{\mathcal{D}}$ uses only the dead-feature rows $\mathbf{W}_{\text{dec}}^{\mathcal{D}}$ of the decoder, with $\mathbf{Z}_{\text{aux}}$ retaining the top-k_{aux} pre-BatchTopK activations within $\mathcal{D}$ per token, where

$$k_{\text{aux}} = \min\big(\lfloor d_{\text{model}}/2 \rfloor,\ |\mathcal{D}|\big), \tag{14}$$

yielding a maximum of 512 active auxiliary features for $d_{\text{model}} = 1024$. The scalar $\alpha = \min(|\mathcal{D}|/512,\ 1)$ adaptively reduces the auxiliary signal when few features are dead, so it does not dominate once the dictionary is healthy. By directing gradient signal through otherwise inactive dictionary entries, this mechanism prevents feature collapse under

heterogeneous training data, where a feature relevant to a minority dataset may stay inactive for many batches dominated by the majority datasets.

At inference, `encode()` applies per-sample top-k instead of batch-level selection, so feature extraction is independent of batch composition and yields deterministic codes per image.

3.5 Training

The four datasets are highly imbalanced: CheXpert and RSNA each contribute approximately 490,000 tokens at Stage 4 (10,000 images $\times$ 49 patches), SIIM-COVID19 contributes $\approx$310,000 tokens, and TB Shenzhen contributes only $\approx$32,000 tokens – a 15$\times$ imbalance between the largest and smallest dataset. Without intervention, gradient updates would be dominated by CheXpert and RSNA, and the SAE would converge to a dictionary describing primarily those datasets' activation structure.

A `StratifiedBatchSampler` enforces strict 25% per-dataset batch composition at every gradient step: exactly $B/4 = 512$ tokens are drawn from each dataset per batch, with the TB Shenzhen corpus sampled with replacement to achieve the required count.

Table 2 reports the key training configurations.

Hyperparameter	Value	Note
d_{model}	1,024	Stage 4 output dimension
d_{SAE}	4,096	Expansion factor $4\times$
k	48	Active features per token (batch-level avg.)
Batch size	2,048	Tokens/step; 512 from each dataset
Training steps	200,000	$\approx$410M token-steps total
n_{dead}	25	Raised from BatchTopK default of 10 to retain minority-dataset features

Table 2: SAE training hyperparameters

3.6 Universality Evaluation Pipeline

The trained SAE is evaluated through a nine-step hierarchical pipeline. Each step gates the next: a latent must pass every prior filter before more expensive tests are applied.

Step 0 – Scanner Latent Audit. Before analysing, latents that encode acquisition identity rather than pathology are identified and excluded. A multinomial logistic regression probe is fitted per latent, predicting dataset identity (CheXpert/RSNA/SIIM/TB) from the latent's per-token activation magnitude. A balanced subsample of 25,000 tokens per dataset (100,000 total) is drawn once and reused for all latent probes to avoid size bias from the larger datasets. The metric is macro-averaged one-vs-rest AUROC; latents with AUROC > 0.80 are flagged as scanner latents and excluded from all subsequent analysis, as an AUROC at this level implies the latent encodes systematic acquisition differences rather than pathology.

Step 1 – Level 1: Activation Frequency. For each unflagged latent i and each dataset d, the activation frequency is computed as:

$$f(i,d) = \frac{|\{t \in d : z_{t,i} > 0\}|}{|d|} \tag{15}$$

A latent is a *universal candidate* if $f(i,d) \geq 0.01$ on at least two of the four datasets. The 1% threshold corresponds to the target sparsity level for TopK SAEs [13]: features below 0.01% are in the dead or noise band; features above 10% encode generic anatomical structure; features at 1% represent the interpretable medium-density band where disease-specific features are expected.

Because each image produces $n_{\text{patches}} = 49$ patch-level codes, patch codes are aggregated per image by taking the element-wise maximum over patches:

$$\tilde{z}_{j,i} = \max_{p=1,\ldots,n_{\text{patches}}} z_{j,p,i} \tag{16}$$

where j indexes images and p indexes spatial patches, yielding one d_{SAE}-dimensional feature vector per image. When label CSVs are supplied, the canonical class-match gate is additionally enforced: the dominant Fleischner group among the top-50 images by max activation must agree across all active datasets. Images absent from the label CSV are excluded from this computation rather than defaulted to `no_finding`.

Step 2 – Level 2: Representational Direction Alignment. For each Level 1 candidate, per-dataset activation centroids are computed in d_{model} space as the mean input vector over all tokens where the latent fires:

$$\boldsymbol{\mu}_{i,d} = \frac{1}{|\{t \in d : z_{t,i} > 0\}|} \sum_{\substack{t \in d \\ z_{t,i} > 0}} \mathbf{x}_t \tag{17}$$

A candidate is *confirmed universal* if every pair of per-dataset centroids satisfies:

$$\cos(\boldsymbol{\mu}_{i,d_1}, \boldsymbol{\mu}_{i,d_2}) \geq 0.70 \quad \forall\, d_1 \neq d_2 \tag{18}$$

The threshold of 0.70 is derived from the SVCCA subspace similarity criterion of $\bar{\rho} \geq 0.50$ [35], raised to 0.70 because this test compares individual directions rather than subspaces. This step excludes a failure mode that frequency alone cannot detect: a latent could clear the 1% threshold on two datasets while responding to entirely different visual patterns in each, as direction alignment would then produce low cosine similarity.

Step 3 – Cross-Dataset Probe. For each confirmed latent, a logistic regression probe [2] is trained on canonical group labels from one dataset and evaluated on each remaining dataset independently, with macro-averaged one-vs-rest AUROC reported per train/test pair. This step is supplementary rather than a pass/fail gate: high cross-dataset AUROC independently validates that the encoder places the same pathology concept in a consistent representational direction across datasets.

Step 4 – Spatial Grounding. For confirmed latents, the Pearson correlation between per-patch activation magnitude and patch-level pathology coverage is computed using radiologist bounding boxes from RSNA and SIIM-COVID19:

$$r_i = \text{Pearson}(\{z_{t,i}\}_{t \in \text{bbox-annotated}},\ \{o_t\}_{t \in \text{bbox-annotated}}) \tag{19}$$

where $o_t \in [0, 1]$ is the fractional area of patch t covered by at least one bounding box. Significance is assessed after Bonferroni correction: $\alpha_{\text{pooled}} = 0.05/n_{\text{confirmed}}$.

Step 5 – Split Detection. Feature splitting is detected by computing the Jaccard similarity of top-50 activating image sets across all pairs of confirmed latents:

$$J(i, j) = \frac{|\text{top}_K(i) \cap \text{top}_K(j)|}{|\text{top}_K(i) \cup \text{top}_K(j)|} \tag{20}$$

Pairs with $J(i, j) \geq 0.50$ are flagged as potential feature splits, where a single clinical concept has been distributed across multiple SAE latents each firing on a dataset-specific subset.

Step 6 – Null Baseline. Both Level 1 and Level 2 tests are applied to two null models on the same activation data. The **random SAE** shares the trained architecture ($d_{\text{model}} = 1024$, $d_{\text{SAE}} = 4096$, $k = 48$) but uses randomly initialised weights; it is expected to pass Level 1 at a high rate (dense random projections fire broadly) but Level 2 at a low rate (random directions are not geometrically aligned across datasets). The **PCA baseline** uses the top-d_{SAE} principal components of the pooled activation data fitted via truncated SVD, with each component treated as a latent that fires when its projection magnitude exceeds the top-k per-sample threshold; this establishes whether the SAE provides representational decomposition beyond what linear dimensionality reduction finds in the encoder representations.

Step 7 – Manual Inspection. For each confirmed latent, the top-8 activating images per dataset are retrieved and displayed with the 7×7 activation heatmap overlaid on the original image, following the manual feature-inspection protocol established for SAEs in language models [7, 45]. This step serves two functions: it assigns semantic labels to the confirmed latents, and it provides the qualitative spatial-grounding track complementary to Step 4. A latent whose patch-grid maxima concentrate on the same anatomical region across all four datasets is judged spatially coherent under manual inspection, including in cases where no bounding-box annotation is available for the relevant region. Manual inspection in this thesis was performed by the author and is acknowledged as a non-blinded single-rater assessment.

A complementary deletion-mode ablation tests which patches the latent causally depends on. For the highest-activating image per dataset, each of the 49 patches is masked in turn and the activation re-encoded through the SAE; the drop in the latent's max-pooled scalar (the *deletion-delta*) is recorded for each patch and arranged as a 7×7 map. Large positive deltas mark causally load-bearing patches; near-zero deltas mark incidental ones. Whereas the patch-grid view shows *where* a latent attends across many images, the ablation shows *which* patches it depends on within one. Deletion-delta heatmaps appear as the right column of each panel in Section 4.6.3.

Step 8 – Illness-Stratified Clinical Characterisation. To determine whether confirmed latents encode clinically meaningful concepts rather than generic anatomical signals, each latent is evaluated through a post-hoc illness-stratified analysis. Images are grouped by

canonical Fleischner illness category, and three metrics are computed. The **pathology-preference ratio** ρ_k:

$$\rho_k = \frac{\frac{1}{6}\sum_{g \neq 5} M_{g,k}}{M_{5,k}} \tag{21}$$

where $M_{g,k}$ is the mean image-level max-pooled activation of latent k on illness group g averaged across datasets and group $g = 5$ is `no_finding`, quantifies whether a latent prefers pathological images over healthy ones. A **discriminability probe** fits a logistic regression with one-vs-rest decomposition using the scalar per-image max-pooled activation as the sole feature, evaluated by 5-fold stratified cross-validation macro-averaged AUROC [2]. **Illness-stratified spatial grounding** computes the Pearson correlation between per-patch activation and bounding-box coverage separately within each illness group, which is strictly stronger evidence than pooled spatial grounding as it tests whether the latent fires at annotated lesion locations specifically within the sick population.

The combination of ρ_k, AUROC, and spatial grounding classifies each confirmed latent into one of four tiers along the structural-to-pathological axis: (1) **structural / background** ($\rho_k < 2$); (2) **mild pathology preference** ($2 \leq \rho_k < 4$); (3) **moderate pathology preference** ($4 \leq \rho_k < 7$); and (4) **strong pathology preference** ($\rho_k \geq 7$). The single-feature discriminability AUROC and pooled spatial r are reported alongside as continuous secondary characterisations rather than as tier thresholds.

4 Evaluation of the Investigation

4.1 Evaluation Criteria

The universality pipeline defined in Section 3.6 imposes a hierarchy of conditions: cross-dataset frequency (L1), representational direction alignment with class match and split-pair monosemanticity (L2), and Bonferroni-corrected spatial grounding against radiologist bounding boxes (L3). A latent is counted as *confirmed universal* once it passes L1 and L2; L3 is reported as the strictest additional criterion, distinguishing the spatially grounded subset within the confirmed set. A sound evaluation must additionally show that confirmed latents are not artefacts of scanner or acquisition differences, that they exceed what is expected by chance or by linear dimensionality reduction, and that they encode clinically meaningful concepts rather than generic anatomical signals. The remainder of this section presents these analyses in turn.

4.2 SAE Training Convergence

The SAE converged to a regime sufficient to ground the analyses below. Over 200,000 training steps (Section 3.5), explained variance (EV) rose from 0.006 at step 1 to 0.934 by step 100,000 and improved by under 2% over the final 100,000 steps. The dead-latent fraction peaked at 0.54 near step 1,000, fell to zero by step 10,000, and remained at zero for the rest of training, confirming that the auxiliary revival loss prevented permanent collapse. The final EV of 0.934 at $k = 48$ active latents per token matches the reconstruction quality reported for BatchTopK SAEs on language-model residual streams [8], supporting the interpretation of the dictionary as a faithful sparse decomposition of UMedPT's Stage 4 activations.

4.3 Scanner Latent Audit

The dictionary does not encode acquisition source as a primary feature. The dataset-identity probe (Section 3.6) flagged zero of the 4,096 latents at the macro-AUROC > 0.80 threshold; the per-latent AUROC distribution was tightly centred near 0.50, consistent with random-chance discrimination of dataset origin. All 4,096 latents proceeded to universality evaluation without exclusion, indicating that the stratified sampling and RMS normalisation pipeline (Section 3.5) suppressed dataset-level magnitude differences in the encoder's activation space.

4.4 Universality Pipeline Results

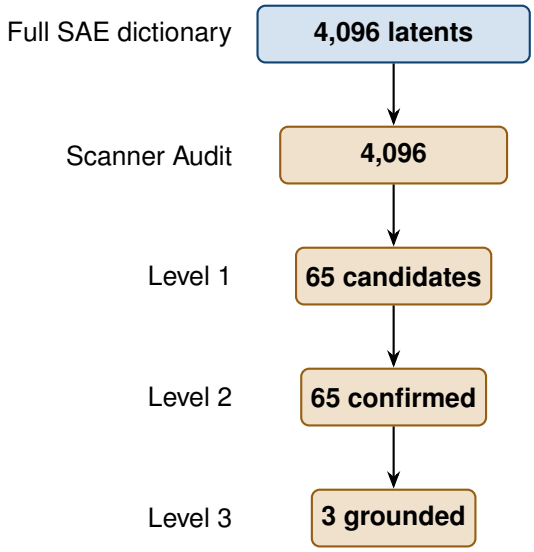


Figure 4: **Three-level universality pipeline applied to Stage 4 latents.**

4.4.1 Level 1: Activation Frequency Gate

Sparse cross-dataset activation isolates a small candidate set. Of 4,096 dictionary latents, 65 (1.59%) activated on more than 1% of patch tokens in at least two datasets; all 65 reached the threshold exclusively on CheXpert and RSNA Pneumonia, confirming the scope statement of Section 3 that SIIM-COVID19 and TB Shenzhen serve as auxiliary rather than primary evidence. Activation frequencies for the 65 candidates ranged from 1.00% to 3.97% on CheXpert (mean 1.75%) and from 1.00% to 3.12% on RSNA (mean 1.38%), within the sparse regime expected for features encoding specific anatomical or pathological concepts.

4.4.2 Level 2: Representational Direction Alignment

All 65 candidates encode stable directions in activation space. Cosine similarity between per-dataset activation centroids in d_{model} space ranged from 0.790 to 0.984 (median 0.957, mean 0.949), with all values exceeding the 0.70 threshold. Because Level 1 already required firing on two or more datasets, the 100% Level 2 retention reflects in part this upstream constraint and should be read as a geometric refinement of the frequency gate rather than an independent confirmation. Zero split pairs were detected by the Jaccard

test, consistent with monosemantic encoding rather than the two-meaning superposition signature of polysemantic features [12]. Cosine alignment is a geometric property and does not guarantee identical semantic content across datasets; the clinical characterisation in Section 4.6 provides the complementary semantic evidence through pathology preference ratios and discriminability probes. All 65 latents matched on the `no_finding` canonical class because it is the most prevalent label across CheXpert and RSNA and consequently appears in nearly every top-50 set; the gate therefore confirms cross-dataset co-activation consistency but does not by itself establish pathology-specific semantics, which the ρ_k ratio and discriminability probe in Section 4.6 address directly.

4.4.3 Level 3: Spatial Grounding

Spatial grounding is assessed along two complementary tracks. The first is the Bonferroni-corrected Pearson correlation between per-patch latent activation and radiologist bounding-box overlap reported in this section. The second is qualitative manual inspection of patch-grid activations and deletion-mode ablation maps reported in Section 4.6.3, following the manual feature-inspection protocols established in the SAE literature on language models [7, 45]. The two tracks answer different questions: the Pearson test asks whether activation magnitude correlates with annotated pathology coverage across all patches and is therefore restricted to the two datasets with bounding boxes (RSNA, SIIM-COVID19); manual inspection asks whether the spatial distribution of activation is anatomically coherent within and across all four datasets, including in regions (mediastinum, diaphragm, cardiac silhouette) for which no bounding-box ground truth exists.

Two properties of the Pearson metric should be noted before the numbers. First, the correlation is computed over all 49 patches per image, the great majority of which contain no pathology; this dilutes any signal localised to a small lesion area, attenuating absolute correlation magnitudes regardless of how cleanly the latent localises. Second, the Bonferroni correction is applied across all 65 confirmed latents ($\alpha' = 7.69 \times 10^{-4}$), a strict threshold relative to the per-test $\alpha = 0.05$. Reported r-values should therefore be read as statistically significant but heavily attenuated localisation signals, not as strong correlations.

Under these conditions, three latents passed Bonferroni correction:

- **Latent 264**: $r = 0.118$, $p = 3.1 \times 10^{-33}$, cosine 0.977
- **Latent 2694**: $r = 0.090$, $p = 4.4 \times 10^{-20}$, cosine 0.971
- **Latent 3868**: $r = 0.047$, $p = 1.8 \times 10^{-6}$, cosine 0.983

Spatial co-localisation under either track is correlational evidence and does not establish that the model causally uses these latents to predict disease [14].

Latent 264 is the strongest of the three. Its top-activating patches across all four datasets concentrate at peripheral lung regions, costodiaphragmatic angles, and pleural boundaries – anatomical loci where radiologist-annotated pathology most frequently occurs. Combined with its Level 2 cosine of 0.977, it provides the clearest single example of a feature whose representational direction *and* spatial grounding both transfer across datasets. Latent 2694 localises to central and perihilar regions consistent with mediastinal or vascular structure; latent 3868 shows the smallest correlation and localises to lower-lobe and diaphragmatic patches. A fourth latent, 2257, produces deletion-mode ablation maps consistently centred on the mediastinal region across all four datasets but fails the formal test because cardiac

and mediastinal regions lack pathology bounding boxes in CheXpert and RSNA – a coverage gap rather than a localisation failure. The patch-grid panels for latents 264, 589, and 2257 appear in Figure 5 (Section 4.6.3).

4.5 Null Baseline Comparison

The trained dictionary captures cross-dataset structure that linear methods do not, while remaining selective relative to random projections. Following the protocol of Karvonen et al. [21], the comparison yielded the analysis presented in Table 3.

Model	d_{SAE}	Level 1 candidates	Level 2 confirmed
Trained BatchTopK SAE	4,096	65	65
Random SAE	4,096	1,569	1,490
PCA	1,024	238	16

Table 3: Level 2 confirmed latents for the trained SAE and two null baselines.

PCA comparison. PCA components are by construction the directions of maximum variance in the pooled activation corpus and are therefore the strongest lower bound on cross-dataset consistency achievable by linear methods. The trained SAE confirmed 65 latents against PCA's 16 – a 4.06-fold absolute excess – with both methods confirming a comparable fraction of their dictionaries (1.59% vs. 1.56%). The absolute gap shows that sparse dictionary learning recovers cross-dataset structure that the orthogonal basis of maximum-variance directions does not capture.

Random SAE. The random SAE confirmed 1,490 latents, more than the trained model. This reflects the Johnson-Lindenstrauss property of high-dimensional random projections [19]: arbitrary directions can produce spurious cosine alignment between per-dataset centroids without encoding any specific concept. The random baseline therefore acts as an upper-bound pathology check – a well-trained SAE should produce *fewer* confirmations than a random one because it allocates dictionary capacity to selective rather than diffuse features. The trained SAE's count of 65, well below the 1,490 random ceiling and well above the 16 PCA floor, is consistent with selective, learned cross-dataset structure rather than random alignment or generic principal directions.

4.6 Clinical Characterisation of Confirmed Latents

4.6.1 Pathology Preference Ratio

The confirmed dictionary spans the full gradient from anatomy to pathology. The pathology preference ratio ρ_k (Section 3.6) ranges from 0.86 to 10.87 across the 65 confirmed latents, with a median of 3.2: most latents fire moderately more on pathological images, a small tail shows strong pathology preference, and a handful behave as structural or near-uniform background detectors (Table 4).

This breadth indicates that universality in UMedPT does not reduce to a single feature class: confirmed latents range from anatomical landmarks to specific pathology-presence detectors, supporting the hypothesis that the encoder represents clinical structure as a graded dictionary rather than a binary disease/no-disease axis.

Latent	Characterisation	ρ_k
3797	Strong illness preference	10.87
1088	Strong illness preference	9.75
3488	Strong illness preference	8.15
694	Illness preference	7.79
589	Illness preference	6.96
784	Near-uniform	1.03
4087	Near-uniform / inverse	1.02
981	Slight normal preference	0.92
380	Slight normal preference	0.91
2102	Slight normal preference	0.86

Table 4: Extremes of the pathology preference ratio ρ_k among the 65 universal latents (top: highest; bottom: lowest).

4.6.2 Pathology Discriminability Probe

Three confirmed latents carry pathology-relevant signal that generalises to all four datasets simultaneously. Macro-AUROC of the single-feature discriminability probe (Section 3.6) ranged from 0.47 to 0.67 across the 65 confirmed latents (median 0.54) – the regime expected for monosemantic features, where each latent encodes one specific concept and a single concept does not fully separate disease from normal. The hypothesis-relevant question is therefore not whether single latents are strong classifiers but whether any latent's signal recurs consistently across *all* datasets. Three did, with 5-fold stratified CV macro-AUROC on CheXpert / RSNA / SIIM-COVID19 / TB Shenzhen:

- **Latent 4087**: 0.529 / 0.666 / 0.608 / 0.585
- **Latent 2257**: 0.569 / 0.630 / 0.560 / 0.579
- **Latent 1576**: 0.533 / 0.621 / 0.611 / 0.578

Latent 2257 is the only confirmed latent with AUROC above 0.55 on every dataset, distinguishing it as the most consistently discriminative single feature in the dictionary; absolute discriminability remains modest (AUROC 0.56–0.63). Latent 4087 ($\rho_k = 1.02$) operates by inverse mechanism: it fires uniformly across normal radiographs and is suppressed on pneumonia, discriminating disease by *absence* rather than presence – consistent with encoding a generic frontal chest geometry disrupted by consolidation or effusion. Together these three provide the most direct empirical support for within-model universality of pathology-relevant signal: a feature whose discriminative behaviour holds across patient populations, institutions, and disease contexts cannot be a dataset-specific shortcut.

The fine-grained probe (14 CheXpert columns, 1 RSNA, 4 SIIM) produced 1,170 NaN cells out of 1,235 because most rare-pathology labels have fewer than the 10 positive examples per fold required for stratified cross-validation; the valid fine-grained results were dominated by RSNA pneumonia and were numerically identical to the coarse probe.

4.6.3 Manual Spatial Inspection: Feature Gallery

The patch-grid activations and deletion-mode ablation maps reported below constitute the qualitative spatial-grounding track introduced in Section 4.4. Manual feature inspection of

this form is the standard validation procedure in the SAE literature [7, 45]; agreement is judged here on two criteria: (i) consistency of the activated region across all four datasets, and (ii) anatomical coherence of that region with a recognisable structure (lung field, diaphragm, mediastinum, costophrenic angle).

Three exemplar latents illustrate the range of universal concepts encoded by the confirmed dictionary, drawn from distinct universality criteria (Table 5). All three satisfy L1 and L2; only latent 264 passes L3, the optional Bonferroni-corrected spatial test, because the remaining two localise to diaphragmatic or central thoracic regions not annotated as pathology bounding boxes in CheXpert or RSNA. They are visualised in Figure 5, each representing a different criterion: spatial grounding (264), anatomical-landmark encoding combined with pathology preference (589), and cross-dataset discriminability (2257).

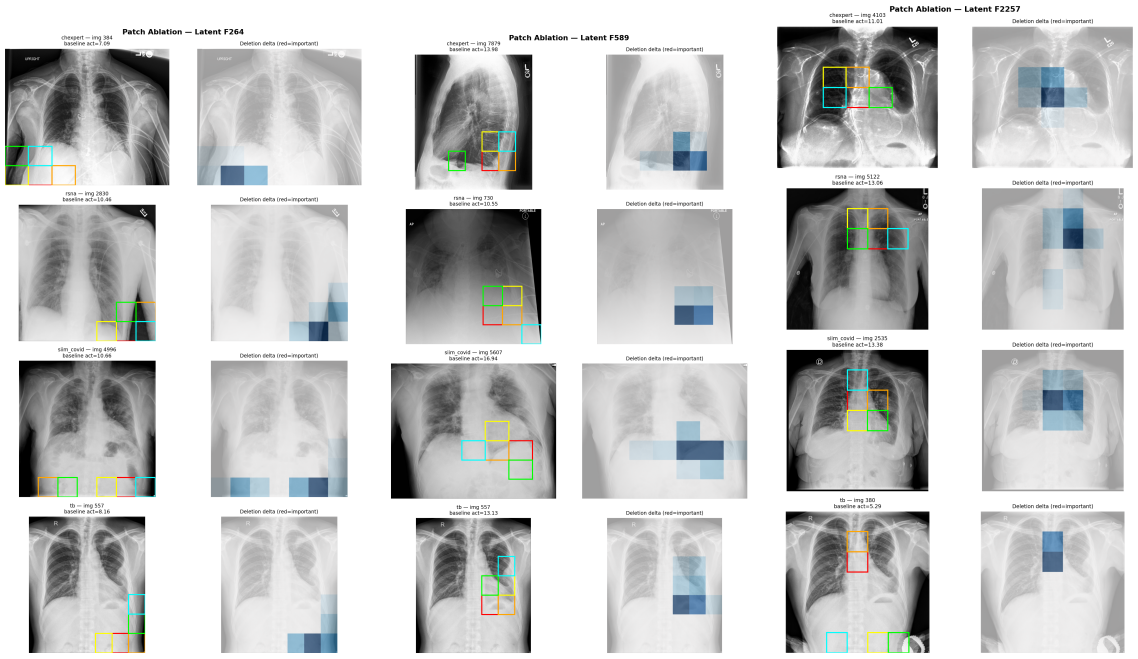
Latent	L1 freq $\geq 1\%$ (CheXpert/RSNA)	L2 cosine ≥ 0.7	L2 class match	L3 spatial Bonferroni ($p < 7.69 \times 10^{-4}$)
F264	1.15% / 1.11%	0.977	Yes	Yes, $p = 3.1 \times 10^{-33}$
F589	1.63% / 1.47%	0.982	Yes	supplementary (no bbox match)
F2257	1.13% / 1.40%	0.961	Yes	supplementary (no bbox match)

Table 5: Universality compliance for the three exemplar latents.

Latent 264 (spatially grounded, costodiaphragmatic angle). The strongest of the three Bonferroni-passing latents (Section 4.4); its deletion-mode patch ablation appears as panel (a) of Figure 5.

Latent 589 (anatomical landmark with high pathology preference, $\rho_k = 6.96$). Latent 589 represents a mode of universality distinct from the spatially grounded latents. Its top-activating patches consistently identify the superior aspect of the diaphragm across all four datasets and remain robust across posteroanterior and lateral acquisitions, suggesting an anatomical landmark rather than a view-dependent pixel pattern. The latent achieves the highest cross-dataset centroid cosine in the confirmed set (0.982), single-latent AUROC above 0.50 on every dataset (CheXpert 0.578, RSNA 0.541, SIIM-COVID19 0.561, TB Shenzhen 0.549), and $\rho_k = 6.96$. The combination of anatomical specificity, cross-view robustness, and pathology preference distinguishes it from both the spatially grounded latents above and the narrow pathology detectors below. Panel (b) of Figure 5.

Latent 2257 (cross-dataset discriminative, central thoracic). The only confirmed latent whose single-latent AUROC exceeds 0.55 on every dataset simultaneously. Its deletion-mode ablation maps concentrate consistently over the central thoracic region (mediastinum and adjacent thoracic spine); the co-localisation does not pass the formal Bonferroni test because the relevant region lacks bounding-box annotation in CheXpert and RSNA. Panel (c) of Figure 5.



(a) Latent 264 - costodiaphragmatic

(b) Latent 589 - diaphragm

(c) Latent 2257 - central thoracic region

Figure 5: **Deletion-mode patch ablation maps for three exemplar confirmed universal latents**

Two further high-ρ_k latents extend the dictionary's coverage of pathology-presence concepts but are not visualised. **Latent 3797** ($\rho_k = 10.87$, the highest in the dictionary) functions as a narrow lung-base / costophrenic-angle airspace-opacity detector with single-latent AUROC of 0.55 on CheXpert. **Latent 1088** ($\rho_k = 9.75$) fires on lower-thoracic regions with an appearance visually distinct from latent 3797's parenchymal pattern, suggesting that the two high-ρ_k latents encode separate pathological concepts rather than a duplicated signal. Together with the visualised exemplars, these latents indicate that the confirmed set spans the full conceptual range from anatomical landmarks to disease-specific pathology presence – consistent with a graded dictionary of plausibly clinically-relevant concepts rather than a narrow set of disease detectors.

5 Conclusions

This thesis investigated whether UMedPT, a multi-task multi-modality medical foundation model, encodes pathological and anatomical concepts that are stable across patient populations and disease contexts within chest radiography. We trained a BatchTopK sparse autoencoder [8] on UMedPT Stage 4 activations harvested from CheXpert, RSNA Pneumonia, SIIM-COVID19, and TB Shenzhen, and evaluated the resulting 4,096-atom dictionary through a nine-step hierarchical pipeline (Section 3.6) that filters latents by frequency, representational direction, spatial grounding, and clinical behaviour, with PCA and random-SAE null comparisons at every stage.

Our experiments returned 65 confirmed universal latents. These fire above 1% on CheXpert and RSNA Pneumonia, hold cross-dataset cosine alignment between 0.79 and 0.98, and span a graded spectrum from anatomical landmarks to specific pathology-presence detectors. Three latents carry the substantive findings. Latent 264 (costodiaphragmatic angles, $r = 0.118$) is the strongest of three to clear Bonferroni-corrected spatial alignment against radiologist bounding boxes. Latent 589 ($\rho_k = 6.96$, cosine 0.982) is the most directionally stable feature in the confirmed set, locked to the superior aspect of the diaphragm across all four datasets and behaving as an anatomical landmark with high pathology preference. Latent 2257 is the only confirmed latent whose single-feature AUROC exceeds 0.55 on every dataset evaluated, the cleanest cross-dataset discriminative signal in the population. Against the null comparisons, the trained dictionary captures $4.06\times$ the cross-dataset structure of PCA and produces 65 confirmations against the 1,490 expected from random projections.

The answer to the research question of Section 1 is therefore a scope-bounded yes. UMedPT's pretraining objective contains no term that ties features together across datasets, yet an unsupervised sparse dictionary recovers 65 internal directions on which the two largest corpora agree, together with a smaller subset of spatially-grounded and manually-inspected latents that represent a recognisable slice of chest-radiograph pathology. BatchTopK is a suitable choice for this regime; the broader claim supported is that SAE-based mechanistic interpretability can be applied to medical foundation models in the first place.

5.1 Further Research

Future work follows directly from the boundaries of this study. Frequency was thresholded absolutely at 1% and that threshold was reached only on the two largest datasets; spatial grounding was measured as a Pearson correlation pooled over 49 patches per image, of which the great majority contain no pathology. A relative, corpus-size-calibrated frequency criterion together with a per-image intersection-over-union spatial metric would loosen both restrictions in one move and re-admit SIIM-COVID19 and TB Shenzhen as primary rather than auxiliary evidence.

More substantively, latents in this thesis are confirmed by their alignment with clinical structure across datasets, not by causal use in the model's predictions. A targeted ablation or activation-steering experiment, run on a downstream classifier built on UMedPT, would close that gap. Beyond that, applying the pipeline to other medical foundation models (RAD-DINO, BiomedCLIP, MedSAM) would isolate findings specific to UMedPT from those generic to the foundation-model paradigm, and applying it to imaging modalities outside chest radiography – with histopathology the harder case, since its diagnostic content is

not legible to a non-specialist – would test whether the cross-dataset features recovered here generalise across medical vision or are bounded by the projection geometry of frontal radiographs.

Whether sparse autoencoders are the right tool for clinical AI auditing is a separate question, and not one this thesis settles. What it does suggest is that the technique survives the move from language models to medical vision, that it surfaces features which are at minimum stable across patient populations and at best interpretable to a non-specialist reader, and that both properties are recoverable from a frozen encoder without task-specific supervision. That is a useful starting point for a problem – understanding what medical foundation models have actually learned – that performance metrics on their own will not solve.

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